Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Systems-focused software engineer with 2+ years building concurrent workflows, autonomous-system simulations, cross-platform software, and real-time integrations. Experienced in C++, Python, state machines, multithreading, closed-loop evaluation, automated testing, performance debugging, and reliable delivery across Windows, Linux, and macOS.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python platform services for 500+ users using Azure SQL and Blob Storage, tuning indexed queries and backend access paths to reduce system response latency from 1.2 seconds to 350 milliseconds.
- Implemented command-style workflows for 5,000+ monthly jobs using explicit state transitions, retries, reconciliation, and failure isolation, tripling throughput while improving runtime correctness, observability, maintainability, and issue investigation.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask REST services handling 10,000+ daily requests, integrating secure ingestion, authentication, job-state updates, and asynchronous grading workflows while investigating production failures affecting 150+ users under peak load.
- Created multithreaded processing for 1,000+ files with validation, progress reporting, shared-state persistence, and failure handling, reducing turnaround 60% while maintaining deterministic behavior under concurrent production load conditions.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible autonomous-system simulation pipelines processing 100,000+ trajectory logs, applying automated evaluation, performance instrumentation, and systematic debugging to improve long-horizon planning accuracy 18% across diverse controlled scenarios.
- Developed a spatial-temporal encoder integrating map geometry and agent state through multi-head attention, improving closed-loop stability 15% while validating control behavior and performance across repeated simulation environments.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed cross-platform Python services and system integrations using AWS Glue, S3, SQL, filtering, and pagination, improving response times 30% and data-processing efficiency 68% across diverse production environments.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating system-level issue diagnosis, release validation, and reliable production deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular Node.js services across eight platform components with authentication, WebSocket messaging, and state synchronization, coordinating 2 million+ Azure SQL records through consistent interfaces and real-time updates.
- Automated Docker builds, functional tests, staged deployments, and runtime monitoring through Jenkins and Azure, reducing release time 50% while improving code quality, system consistency, and operational visibility.

Skills

Languages & Systems: C++, Python, Java, Bash, JavaScript, TypeScript, SQL, object-oriented programming, data structures, algorithms, concurrency, multithreading, state machines
Platforms & Integration: Windows, Linux, macOS, Node.js, Flask, FastAPI, REST APIs, WebSockets, system integration, asynchronous workflows, distributed workflows, spatial modeling
Autonomy & Data: autonomous-system simulation, trajectory processing, closed-loop evaluation, control behavior, PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, AWS RDS, data modeling
Quality & Delivery: Git, Docker, Jenkins, CI/CD, PyTest, Mockito, automated testing, unit testing, integration testing, simulation, benchmarking, debugging, instrumentation, monitoring, code review, failure isolation, Agile

Projects

CPU-Scheduling Simulator | Live | *C++, Simulation, Concurrency, Operating Systems, React*
- Implemented six scheduling algorithms in C++ with deterministic simulations, state transitions, and performance instrumentation, evaluating timing behavior across 1,000+ process configurations while reducing systematic debugging time 20%.

PiXMod | Live | *Next.js, TypeScript, MongoDB, Stripe, CLIP, SAM*
- Built an interactive computer-vision application using CLIP and SAM, implementing TypeScript workflows, reusable interfaces, secure authentication, MongoDB persistence, and responsive image-processing behavior across multiple client device form factors.

FinNexus | Live | *TypeScript, React, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built real-time workflows supporting 50,000+ users and one million transactions, integrating REST APIs, WebSocket messaging, state validation, MongoDB persistence, runtime monitoring, and automated deployment through reliable CI/CD pipelines.